CAST-CD: A Lightweight Cross-Attention Siamese Transformer for Bi-Temporal Remote Sensing Image Change Detection

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***Abstract*—Bi-temporal remote sensing image change detection identifies land-cover changes by comparing images of the same geographical region acquired at different times. Accurate detection remains challenging due to illumination differences, image noise, spectral similarity, and class imbalance. This paper presents CAST-CD, a Cross-Attention Siamese Transformer for Change Detection, designed to balance detection accuracy and computational efficiency. The framework employs a shared-weight Siamese encoder based on selected EfficientNet-B4 layers to extract temporal features. A Global Differential Attention Fusion (GDAF) module emphasizes informative channels in the absolute feature differences, while a Bidirectional Cross-Temporal Attention (CTA) module captures interactions between the two temporal representations. An upsampling decoder with skip connections generates a pixel-level binary change map. Preliminary evaluation on WHU-CD, LEVIR-CD, and DSIFN-CD yields F1-scores of 94.25%, 88.18%, and 91.07%, respectively, with corresponding intersection-over-union scores of 89.12%, 78.86%, and 83.60%. The current implementation is reported to contain 314,531 parameters and require 1.965 GFLOPs for paired 256×256 image inputs. These findings indicate the potential of combining differential attention and explicit temporal interaction within a compact architecture, while differences across datasets highlight the need for further refinement and controlled evaluation.**

***Index Terms*—Bi-temporal change detection, remote sensing, satellite imagery, Siamese networks, cross-temporal attention, lightweight deep learning.**

I. Introduction

Remote sensing image change detection is the process of identifying differences in the state of an object or phenomenon by observing it at different times. Bi-temporal change detection, which relies on comparing two co-registered images acquired over the same geographical area at distinct timestamps, plays a critical role in various real-world applications [1], [24]. These applications range from urban expansion monitoring and deforestation tracking to disaster damage assessment, flood mapping, and agricultural surveys. In these operational scenarios, satellite sensors capture both short-term dynamic events (such as wildfires, inundations, and rapid construction) and long-term evolutionary trends (such as urban sprawl and gradual vegetative depletion). A fundamental challenge in this domain is accurately differentiating genuine semantic changes from irrelevant environmental variations caused by seasonal shifts, illumination differences, sensor noise, atmospheric turbulence, and soil moisture variations [8], [28].

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Historically, classical methods provided the foundational framework for satellite change detection [2], [3]. Early techniques such as direct image differencing, Change Vector Analysis (CVA), and Principal Component Analysis (PCA) were widely adopted due to their computational simplicity and straightforward interpretability [4], [5]. These traditional approaches rely primarily on low-level spectral signatures, manual threshold selection, and handcrafted feature descriptors. However, because they lack high-level semantic awareness, classical techniques are sensitive to illumination inconsistencies, cloud shadows, and radiometric noise [6], [7]. Consequently, although classical methods are computationally lightweight, their limited ability to model complex spatial context restricts their generalisation and detection reliability when applied to modern, high-resolution Earth observation imagery.

To overcome the limitations of handcrafted feature extractors, deep learning methods, predominantly Convolutional Neural Networks (CNNs), have emerged as a widely adopted approach in remote sensing [9], [11]–[14]. Siamese architectures, which employ twin convolutional feature extractors with shared parameter weights, have established a robust framework for processing bi-temporal images symmetrically [16], [18]. Building upon semantic segmentation models, various change detection networks have integrated multi-scale skip connections and dense feature propagation schemes to capture hierarchical representations [17], [31]. Nevertheless, standard convolutional kernels remain constrained by inherently local receptive fields [19]. This spatial locality limits their ability to model long-range spatial context and broader spatial-temporal relationships across high-resolution satellite scenes.

In recent years, attention mechanisms and Vision Transformers (ViTs) have been actively investigated to capture these extensive global dependencies [25], [46]. By converting feature maps into sequence tokens and computing scaled dot-product attention across spatial and temporal dimensions, transformer-based frameworks can model contextual relationships across the image grid [24], [26]. Despite achieving high detection accuracy, conventional vision transformers introduce substantial computational complexity, with quadratic memory requirements with respect to token count and potentially millions of trainable parameters. These resource demands can limit their suitability for edge devices, drone platforms, and large-scale satellite processing pipelines where computational throughput and memory footprints are strictly constrained.

Although transformer-based models can achieve high ac-

curacy, a critical challenge remains in reducing model complexity without compromising the preservation of high-level semantic information and fine-grained spatial details. Many existing lightweight models address this challenge by reducing network depth and model complexity, but often struggle to balance computational efficiency with the preservation of spatial information. This trade-off can result in the loss of fine-grained details, making it difficult for lightweight networks to match the performance of their computationally heavier counterparts on complex urban datasets [37], [38].

A further challenge lies in the effective alignment of temporal features and the precise delineation of change boundaries. Many efficient models rely on simple feature subtraction or concatenation, which may not adequately capture the explicit spatial-temporal interactions required to distinguish subtle changes. Furthermore, these simplified fusion strategies can lead to imprecise boundaries, limiting the accurate delineation of small objects, such as individual buildings, in high-resolution mapping tasks [32], [41]. These limitations highlight the need for lightweight change detection architectures that can explicitly model bidirectional temporal feature interactions while preserving fine-grained spatial information and accurate object boundaries.

To address these challenges, we propose CAST-CD, a Cross-Attention Siamese Transformer for Change Detection. CAST-CD employs a truncated EfficientNet-B4 backbone [45] to extract spatial features. The network incorporates Bidirectional Cross-Temporal Channel Attention (CTA) at the deepest encoder stage. CTA uses learned projections of one temporal representation to construct queries and those of the other temporal representation to construct keys and values, producing channel-wise cross-temporal interactions in both directions. Furthermore, CAST-CD incorporates a Gated Difference-Attention Fusion (GDAF) module with Squeeze-and-Excitation channel weighting [47], an UpConcat decoder with multi-scale skip connections, and a multi-task learning framework comprising boundary supervision, deep supervision [42], and Contrastive Feature Alignment (CFA).

The main contributions of this work are summarised as follows:

1) We introduce CAST-CD, a Siamese change-detection architecture that integrates bidirectional cross-temporal channel attention with a convolutional backbone. The current implementation is reported to require $314,531$ parameters and 1.965 GFLOPs for paired 256×256 inputs.
2) We design a multi-scale feature-interaction framework combining Gated Difference-Attention Fusion (GDAF) at two intermediate encoder stages and Bidirectional Cross-Temporal Channel Attention (CTA) at the deepest stage.
3) We propose a multi-task learning objective incorporating boundary supervision, intermediate deep supervision, and Contrastive Feature Alignment (CFA) to enhance contour delineation and latent feature separability.
4) We conduct empirical evaluations across three benchmark datasets (WHU-CD, LEVIR-CD, and DSIFN-CD), demonstrating competitive accuracy and computational efficiency relative to prominent state-of-the-art baselines.

II. Related Work

A. Siamese Convolutional Architectures

Siamese architectures, originally developed for metric learning and similarity verification, have become a widely adopted paradigm for bi-temporal remote sensing change detection. By utilising two identical sub-networks with shared weights, Siamese models ensure that both temporal images undergo identical feature extraction transformations, projecting them into a shared latent feature space [16], [18]. Early adaptations demonstrated the effectiveness of shared-weight feature extraction for bi-temporal change detection. Subsequent extensions incorporated encoder-decoder designs, leveraging dense nested skip pathways to preserve high-resolution spatial details [17], [31]. Moreover, recurrent and multispectral Siamese formulations [19]–[22] explore intertemporal feature relationships, while other machine learning approaches [10] emphasise robust representation learning for complex Earth observation data. Although these convolutional architectures excel at local feature extraction, conventional convolutions remain constrained by local receptive fields, limiting their ability to capture long-range contextual dependencies across extensive geographical regions.

B. Attention Mechanisms and Transformers

To overcome the limitations of the local receptive fields of standard convolutional filters, attention mechanisms have been widely integrated into remote sensing pipelines [28]–[30], [46]. Channel and spatial attention modules selectively enhance change-informative feature channels and spatial regions while attenuating background clutter. In change detection, spatial-temporal attention networks, such as STANet, introduced dedicated attention modules to capture pixel-level dependencies across space and time [24].

More recently, Vision Transformers (ViTs) have been adopted to model global intertemporal context [25]. The Bitemporal Image Transformer (BIT) converts CNN feature maps into compact semantic tokens and employs transformer layers to model temporal context [25]. ChangeFormer combines a hierarchical transformer encoder with a multi-scale decoder to capture contextual information across different feature resolutions [33]. Dual-branch intertemporal networks, such as DMINet, model multilevel feature differences [26], while Deeply Supervised Attention Metric Networks (DSAMNet) refine feature representations through attention-based feature processing [27]. Similarly, semantic change detection models such as ChangeMask demonstrate the utility of transformer encoders for multi-task change analysis [35]. Despite their ability to model global dependencies, transformer-based architectures can introduce substantial computational overhead due to the quadratic complexity of standard self-attention and their potentially large model footprints. For example, the reported ChangeFormer configuration contains approximately 41.03 million parameters and requires 202.79 GFLOPs, illustrating

the computational demands that can arise in transformer-based change detection [33]. These resource requirements can limit their suitability for edge devices and other resource-constrained applications.

C. Lightweight Models and Feature Fusion

The growing need to deploy change detection models on edge devices and large-scale geospatial computing systems has spurred considerable interest in lightweight network architectures [38], [39]. TinyCD demonstrated that a compact Siamese model containing fewer than $300,000$ parameters could achieve competitive detection accuracy on building change benchmarks [37]. TinyCD leverages low-level feature representations and a spatiotemporal feature-mixing strategy termed the Mix and Attention Mask Block (MAMB). Its reported configuration requires approximately 0.285 million parameters and 3.088 GFLOPs, illustrating the potential for reducing computational complexity while maintaining competitive change detection performance [37].

Other lightweight approaches, such as A2Net, employ a MobileNetV2 backbone with progressive feature aggregation and supervised attention to improve feature representation efficiency [39]. These approaches demonstrate the importance of efficient feature extraction and fusion. Nevertheless, reducing model complexity while preserving high-level semantic information and fine-grained spatial details remains challenging. Feature fusion strategies based primarily on differencing or mixing may not explicitly capture directional relationships between temporal representations, particularly when distinguishing subtle spectral variations from genuine semantic changes in heterogeneous scenes.

D. Multitask Learning and Boundary Supervision

Accurate delineation of building boundaries is essential in high-resolution remote sensing image interpretation [41]. However, lightweight change detection networks can produce imprecise or irregular object boundaries due to aggressive downsampling and the loss of fine-grained spatial details. Multitask learning provides an effective mechanism for addressing this challenge by jointly optimising complementary visual tasks [16], [43]. In remote sensing, multitask frameworks have demonstrated the potential of auxiliary tasks, such as boundary detection and semantic segmentation, to encourage the learning of complementary representations for the primary change detection task. Furthermore, deep supervision techniques, as demonstrated in DSIFN [42], provide direct gradient feedback to intermediate network layers, facilitating the optimisation of deeper architectures. These approaches highlight the importance of incorporating complementary supervision to improve feature learning and spatial delineation. However, maintaining precise object boundaries while limiting computational overhead remains an important consideration for lightweight change detection architectures.

E. Research Gaps and Motivation

The reviewed literature highlights three interrelated challenges in remote sensing change detection. First, transformer-based architectures can capture long-range contextual dependencies but may introduce substantial computational costs, whereas lightweight models must balance computational efficiency with the preservation of semantic information [33], [37]. Second, conventional feature differencing and mixing strategies may not adequately represent directional interactions between bi-temporal features, limiting their ability to distinguish subtle changes from irrelevant spectral variations. Third, aggressive feature compression and downsampling can compromise fine-grained spatial information and boundary delineation [32], [41]. These challenges motivate the development of lightweight change detection architectures that combine efficient feature extraction, explicit cross-temporal interactions, and effective preservation of spatial details.

III. Methodology

A. Overall Architecture

Bi-temporal remote sensing image change detection is formulated as a pixel-wise binary classification task. Let $I_1, I_2 \in \mathbb{R}^{H\times W\times 3}$ denote co-registered RGB images acquired over the same geographic area at times T_1 and T_2. The reference change mask is $Y \in \{0,1\}^{H\times W}$, where $Y_{ij} = 1$ denotes a changed pixel and $Y_{ij} = 0$ denotes an unchanged pixel. The network predicts a change-probability map, as expressed in (1), where f_θ denotes the CAST-CD network, θ represents its trainable parameters, and P is the predicted probability map of spatial dimensions $H \times W$

$$P = f_\theta(I_1, I_2), \qquad P \in [0,1]^{H\times W} \tag{1}$$

During inference, the probability map is converted into a binary change map using the thresholding rule in (2), where $\widehat{Y}_{ij}$ is the predicted label at pixel (i, j), $\mathbb{I}[\cdot]$ is the indicator function, and the implementation uses $\tau = 0.5$

$$\widehat{Y}_{ij} = \mathbb{I}[P_{ij} > \tau] = \begin{cases} 1, & P_{ij} > \tau, \\ 0, & P_{ij} \le \tau \end{cases} \tag{2}$$

CAST-CD comprises a shared-weight Siamese encoder based on selected EfficientNet-B4 layers, an initial fusion block for the two input images, GDAF modules at two intermediate encoder stages, a CTA module at the deepest encoder stage, and an UpConcat decoder with skip connections and auxiliary prediction heads. The two temporal images are processed by the same encoder. The intermediate features are recalibrated by GDAF, while CTA models cross-temporal channel relationships at the deepest stage. The decoder combines these multi-scale representations to produce the pixel-level change map. Figure 1 summarizes the main components.

B. Shared-Weight Siamese Encoder

The encoder extracts corresponding feature representations from the two temporal inputs using shared parameters. Let E_{θ_e} denote the pretrained EfficientNet-B4 feature extractor [45]. The encoder outputs at each retained stage are defined in (3), where $E^{(k)}_{\theta_e}$ denotes the shared pretrained EfficientNet-B4 feature extraction operation at stage k, I_1 and I_2 are the two

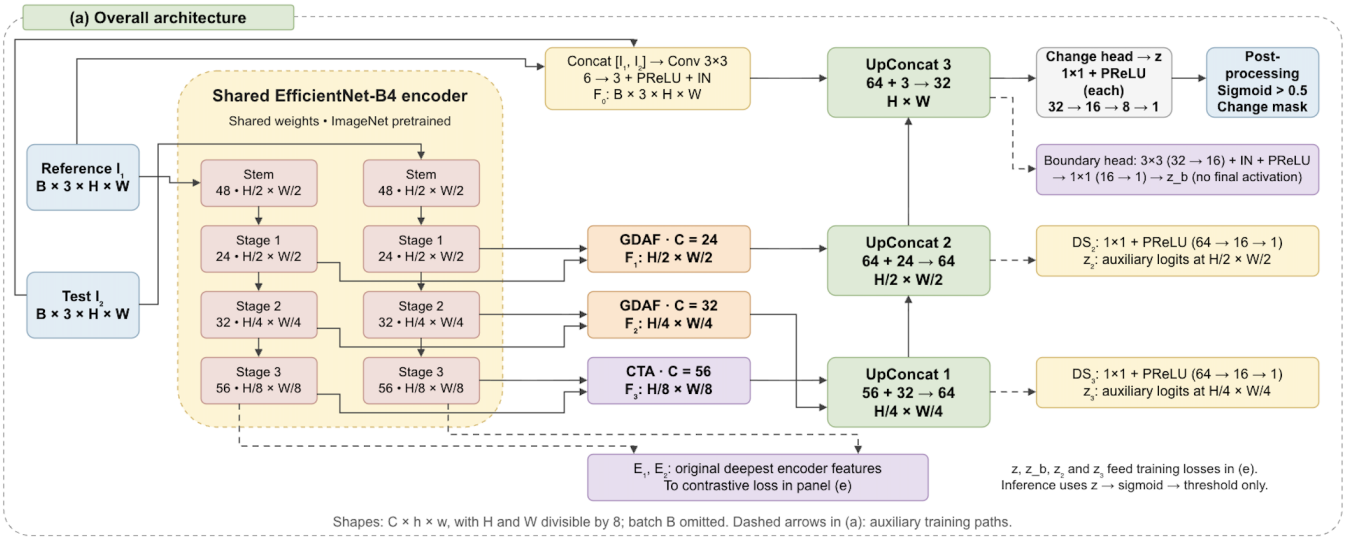


Fig. 1. Overall architecture of CAST-CD. A shared-weight EfficientNet-B4 encoder extracts bi-temporal features. GDAF fuses features at two intermediate scales, while bidirectional CTA processes the deepest features. Three UpConcat blocks reconstruct the full-resolution representation. The change head predicts the change map, while boundary and deep-supervision heads provide auxiliary training supervision.

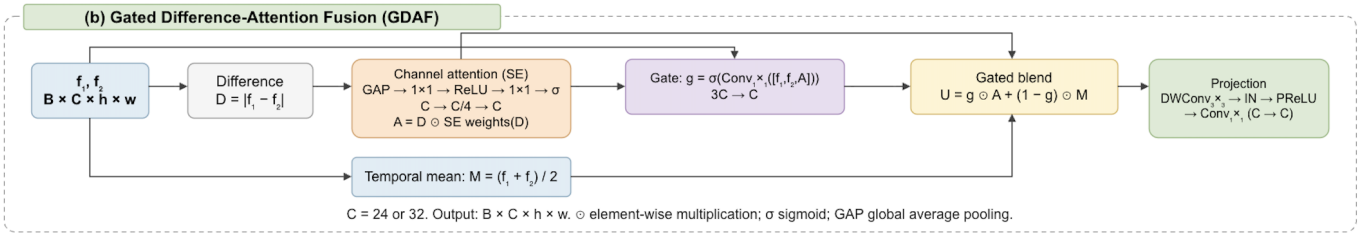


Fig. 2. Gated Difference-Attention Fusion (GDAF). Squeeze-and-excitation weights recalibrate the absolute temporal feature difference. A learned gate blends the attended difference with the mean temporal features. A depthwise convolution, instance normalization, PReLU, and pointwise convolution produce the fused output while preserving spatial resolution and channel count.

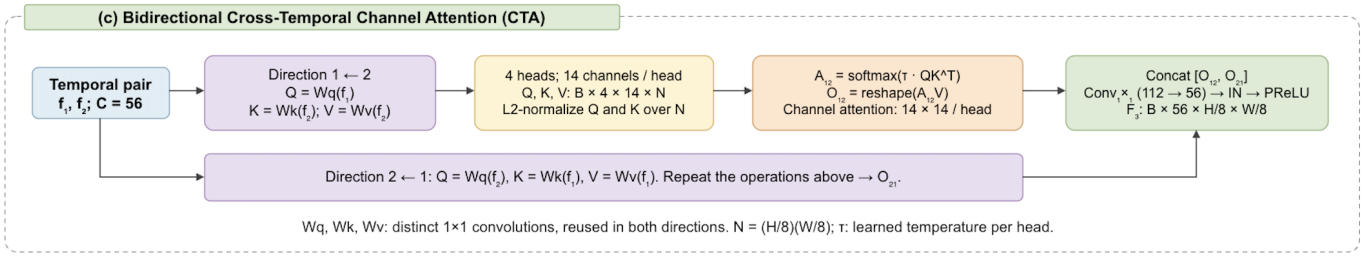


Fig. 3. Bidirectional cross-temporal channel attention (CTA). Queries from one temporal feature map interact with keys and values from the other using four attention heads. Queries and keys are L2-normalized over spatial positions. The two directional outputs are concatenated and projected using a 1×1 convolution, instance normalization, and PReLU.

temporal inputs, and $F_1^{(k)}$ and $F_2^{(k)}$ are their corresponding feature maps

$$F_1^{(k)} = E_{\theta_e}^{(k)}(I_1), \qquad F_2^{(k)} = E_{\theta_e}^{(k)}(I_2) \tag{3}$$

The implementation uses the pretrained backbone through feature layer 3 and does not freeze its parameters. The three retained feature maps have 24, 32, and 56 channels, respectively. GDAF is applied to the first two feature stages, while CTA is applied to the 56-channel deepest representation. The decoder also receives a shallow representation produced by concatenating the two RGB inputs and projecting the resulting six-channel tensor to three channels. Sharing the encoder weights ensures that both temporal images undergo the same feature-extraction transformations.

C. Gated Difference-Attention Fusion

GDAF highlights differences between the two temporal feature maps and applies Squeeze-and-Excitation (SE) channel recalibration [47]. At each of the two intermediate encoder stages, the absolute feature difference is computed as ex-

pressed in (4), where $F_1, F_2 \in \mathbb{R}^{h\times w\times C}$ denote the corresponding temporal feature maps and D is their element-wise absolute difference

$$D = |F_1 - F_2| \tag{4}$$

Global average pooling is applied to D to obtain the channel descriptor in (5), where D_{ijc} is the difference-feature value at spatial location (i, j) and channel c, and z_c is the pooled descriptor for channel c

$$z_c = \frac{1}{hw}\sum_{i=1}^{h}\sum_{j=1}^{w} D_{ijc} \tag{5}$$

The channel weights are calculated as expressed in (6), where W_1 and W_2 denote the learned projection weights, $r = 4$ is the reduction ratio, $\delta(\cdot)$ denotes ReLU activation, and $\sigma(\cdot)$ denotes sigmoid activation. The attended difference is obtained as expressed in (7), where s contains the channel-wise weights and $\odot$ denotes channel-wise multiplication. The learned gate is calculated as expressed in (8), where $[\cdot;\cdot;\cdot]$ denotes channel-wise concatenation, $\mathrm{Conv}_{1\times1}$ denotes a 1×1 convolution, and G is the learned gate.

$$s = \sigma\left(W_2\delta(W_1 z)\right) \tag{6}$$

$$D_{\mathrm{att}} = s \odot D \tag{7}$$

$$G = \sigma\left(\mathrm{Conv}_{1\times1}\left([F_1; F_2; D_{\mathrm{att}}]\right)\right) \tag{8}$$

The gated representation is calculated using (9), where F_{mix} blends the attended difference and the mean temporal feature representation according to G. The final GDAF output is defined in (10), where $\mathrm{Proj}(\cdot)$ denotes the projection block applied to the gated representation. The projection block consists of a 3×3 depthwise convolution, instance normalization, PReLU activation, and a 1×1 pointwise convolution. This operation recalibrates differential features and blends them with the mean temporal representation while preserving spatial resolution. Figure 2 illustrates the GDAF module.

$$F_{\mathrm{mix}} = G \odot D_{\mathrm{att}} + (1 - G) \odot \frac{F_1 + F_2}{2} \tag{9}$$

$$F_{\mathrm{GDAF}} = \mathrm{Proj}(F_{\mathrm{mix}}) \tag{10}$$

D. Bidirectional Cross-Temporal Channel Attention

CTA models cross-temporal relationships at the deepest encoder stage. Unlike spatial-token attention, the implemented module computes attention across channel representations after aggregating information over spatial positions. Let $F_1, F_2 \in \mathbb{R}^{B\times C\times h\times w}$ be the two deepest feature maps, $N = hw$ the number of spatial positions, $M = 4$ the number of attention heads, and $d = C/M$ the channels per head. Learned 1×1 convolutions project each temporal feature map to query, key, and value tensors. After reshaping, these tensors have dimensions $B\times M\times d\times N$. The query and key tensors are L2-normalized along the spatial dimension. For the direction from T_1 to T_2, the channel-attention matrix is calculated as expressed in (11), where $\overline{Q}_1$ and $\overline{K}_2$ are the spatially normalized query and key tensors, respectively, and η is the learned temperature parameter for each head

$$S_{12} = \mathrm{softmax}\left(\eta\,\overline{Q}_1\overline{K}_2^{\top}\right) \tag{11}$$

The two directional attention outputs are defined in (12), where V_1 and V_2 denote the projected value tensors, S_{21} is the reverse-direction attention matrix, and $\mathrm{reshape}(\cdot)$ restores the attention output to the feature-map dimensions

$$A_{12} = \mathrm{reshape}(S_{12}V_2), \qquad A_{21} = \mathrm{reshape}(S_{21}V_1) \tag{12}$$

Here, $S_{21} = \mathrm{softmax}(\eta\,\overline{Q}_2\overline{K}_1^{\top})$, where $\overline{Q}_2$ and $\overline{K}_1$ are the normalized reverse-direction query and key tensors. The fused CTA output is expressed in (13), where $[A_{12}; A_{21}]$ denotes channel-wise concatenation and $\Psi(\cdot)$ denotes the output projection block

$$F_{\mathrm{CTA}} = \Psi([A_{12}; A_{21}]) \tag{13}$$

The projection block consists of a 1×1 convolution, instance normalization, and PReLU activation. This bidirectional channel-attention operation enables the deepest representation from each temporal observation to be informed by the other temporal observation. Figure 3 illustrates the CTA module.

E. Feature Fusion and UpConcat Decoder

The implementation combines temporal information across multiple encoder scales rather than merging the GDAF and CTA outputs at a single scale. First, the two RGB inputs are concatenated and projected from six channels to three channels. This operation is expressed in (14), where $[I_1; I_2]$ denotes channel-wise concatenation and $\phi_0(\cdot)$ denotes the initial block comprising a 3×3 convolution, PReLU activation, and instance normalization

$$F_0 = \phi_0([I_1; I_2]) \tag{14}$$

Here, F_0 provides a shallow skip representation. GDAF produces the two intermediate features $F^{(1)}_{\mathrm{GDAF}}$ and $F^{(2)}_{\mathrm{GDAF}}$, while CTA produces the deepest feature F_{CTA}. The three decoder stages are defined in Equations 15, 16, and 17, where Z_3 and Z_2 denote intermediate decoder representations, U is the final decoded feature map, and each $\mathcal{U}_k$ denotes one upsampling-and-fusion block

$$Z_3 = \mathcal{U}_1(F_{\mathrm{CTA}}, F^{(2)}_{\mathrm{GDAF}}) \tag{15}$$

$$Z_2 = \mathcal{U}_2(Z_3, F^{(1)}_{\mathrm{GDAF}}) \tag{16}$$

$$U = \mathcal{U}_3(Z_2, F_0) \tag{17}$$

Each $\mathcal{U}_k$ denotes bilinear upsampling by a factor of two, concatenation with the corresponding skip feature, and a convolutional block comprising depthwise convolution, PReLU, instance normalization, pointwise convolution, PReLU, and instance normalization. The primary head generates the change-probability map as expressed in (18), where $\mathcal{H}_{\mathrm{main}}$ denotes the pixel-wise prediction head, U is the decoded feature map, and σ is the sigmoid activation. During training, a boundary head predicts boundary logits, and two auxiliary heads (DS2 and DS3) provide intermediate deep supervision.

These auxiliary heads are used to calculate the training loss and are not required for inference.

$$P = \sigma\left(\mathcal{H}_{\text{main}}(U)\right) \tag{18}$$

F. Proposed Multitask Training Objective

CAST-CD is trained using binary cross-entropy (BCE), focal-Tversky loss, boundary supervision, two intermediate deep-supervision losses, and Contrastive Feature Alignment (CFA). The overall training objective is defined in (19), where λ_{BCE}, λ_{FT}, $\lambda_{\text{Boundary}}$, λ_{DS2}, λ_{DS3}, and λ_{CFA} are non-negative coefficients that control the contributions of the respective loss components

$$\begin{aligned}\mathcal{L} = \lambda_{\text{BCE}}\mathcal{L}_{\text{BCE}} + \lambda_{\text{FT}}\mathcal{L}_{\text{FT}} \\ + \lambda_{\text{Boundary}}\mathcal{L}_{\text{Boundary}} + \lambda_{\text{DS2}}\mathcal{L}_{\text{DS2}} \\ + \lambda_{\text{DS3}}\mathcal{L}_{\text{DS3}} + \lambda_{\text{CFA}}\mathcal{L}_{\text{CFA}}\end{aligned} \tag{19}$$

The BCE term is computed from the main-head logits using a numerically stable binary cross-entropy-with-logits implementation. For the focal-Tversky loss, the Tversky index is computed from the soft true-positive (TP), false-negative (FN), and false-positive (FP) contributions. The focal-Tversky objective is defined in (20), where α and β control the relative contributions of false negatives and false positives, respectively, γ is the focusing parameter, and ϵ is a smoothing constant

$$\mathcal{L}_{\text{FT}} = (1 - \text{TI})^{\gamma}, \qquad \text{TI} = \frac{TP + \epsilon}{TP + \alpha FN + \beta FP + \epsilon} \tag{20}$$

Here, $TP = \sum_n p_n y_n$, $FN = \sum_n (1 - p_n) y_n$, and $FP = \sum_n p_n (1 - y_n)$, where p_n and y_n denote the predicted change probability and reference label at pixel n, respectively. grid search may be conducted in the future for setting lambda values

Boundary targets are extracted from the binary reference masks using a 3×3 morphological gradient. The boundary loss is calculated by applying binary cross-entropy with logits to the boundary-head predictions and their corresponding boundary targets. For deep supervision, the reference change mask is resized to the spatial resolution of each auxiliary output using nearest-neighbor interpolation, and BCE with logits is applied to the DS2 and DS3 predictions.

CFA is applied to the deepest temporal feature maps to encourage unchanged feature pairs to become similar while separating changed feature pairs. Before computing the feature distances, both temporal feature vectors are L2-normalized along the channel dimension, and the binary change mask is resized to the feature-map resolution using nearest-neighbor interpolation. The CFA objective is defined in (21), where $d_{bij} = \|f_{1,b}^{(i,j)} - f_{2,b}^{(i,j)}\|_2$ denotes the Euclidean distance between the normalized temporal feature vectors at location (i, j) for batch element b, y_{bij} is the corresponding resized binary change label, B is the batch size, h and w are the feature-map dimensions, and m is the contrastive margin. The first term encourages unchanged temporal features to become similar, whereas the second penalizes changed feature pairs whose distance falls below the specified margin.

$$\begin{aligned}\mathcal{L}_{\text{CFA}} = \frac{1}{Bhw} \sum_{b=1}^{B} \sum_{i=1}^{h} \sum_{j=1}^{w} \Big[(1 - y_{bij}) d_{bij}^2 \\ + y_{bij} \max(0, m - d_{bij})^2 \Big]\end{aligned} \tag{21}$$

G. Overall Workflow

The CAST-CD workflow comprises input preprocessing, multi-scale temporal feature interaction, and decoder-based prediction. Let $I_1, I_2 \in \mathbb{R}^{H \times W \times 3}$ denote the co-registered RGB images acquired at times T_1 and T_2, respectively. The images are independently normalized using their image-wise mean and standard deviation. During training, random 256×256 crops are extracted when required. Horizontal and vertical flips, each with a probability of 0.5, and random rotations of $0°$, $90°$, $180°$, or $270°$ are applied consistently to both images and the reference mask $Y \in \{0, 1\}^{H \times W}$. Boundary targets are generated from the binary change masks using a 3×3 morphological gradient.

The paired images are processed by a shared-weight, ImageNet-pretrained EfficientNet-B4 encoder truncated at feature layer 3, producing temporal feature maps $F_1^{(k)}$ and $F_2^{(k)}$ at the retained stages. GDAF operates on the intermediate features by computing their absolute difference, $D = |F_1 - F_2|$, applying Squeeze-and-Excitation (SE) channel recalibration, and using a learned gate to combine the attended difference with the mean temporal representation. Bidirectional Cross-Temporal Attention (CTA) operates on the deepest feature maps, capturing cross-temporal channel relationships in both directions using learned query, key, and value projections. The resulting multi-scale representations are integrated by the decoder through progressive upsampling and skip connections to recover the change map at the input resolution.

The main segmentation head produces the predicted probability map $P = f_\theta(I_1, I_2)$, while the boundary and intermediate deep-supervision heads provide auxiliary supervision during training. The overall objective combines binary cross-entropy (BCE), focal-Tversky loss, boundary supervision, deep supervision, and Contrastive Feature Alignment (CFA). During inference, the probability map is converted into a binary prediction using $\hat{Y}_{ij} = \mathbb{I}[P_{ij} > \tau]$, where $\tau = 0.5$. Performance is assessed using standard change-detection metrics, including precision, recall, F1-score, and intersection over union (IoU).

IV. Experimental Setup

A. Datasets

The performance of CAST-CD is evaluated on three widely recognized remote sensing change detection benchmarks: WHU-CD, LEVIR-CD, and DSIFN-CD. While WHU-CD and LEVIR-CD concentrate on binary building changes under high-resolution aerial and satellite views, DSIFN-CD serves as a challenging multi-class land-cover change benchmark characterized by diverse spatial and spectral dynamics. Representative samples from each dataset are illustrated in Figure 4.

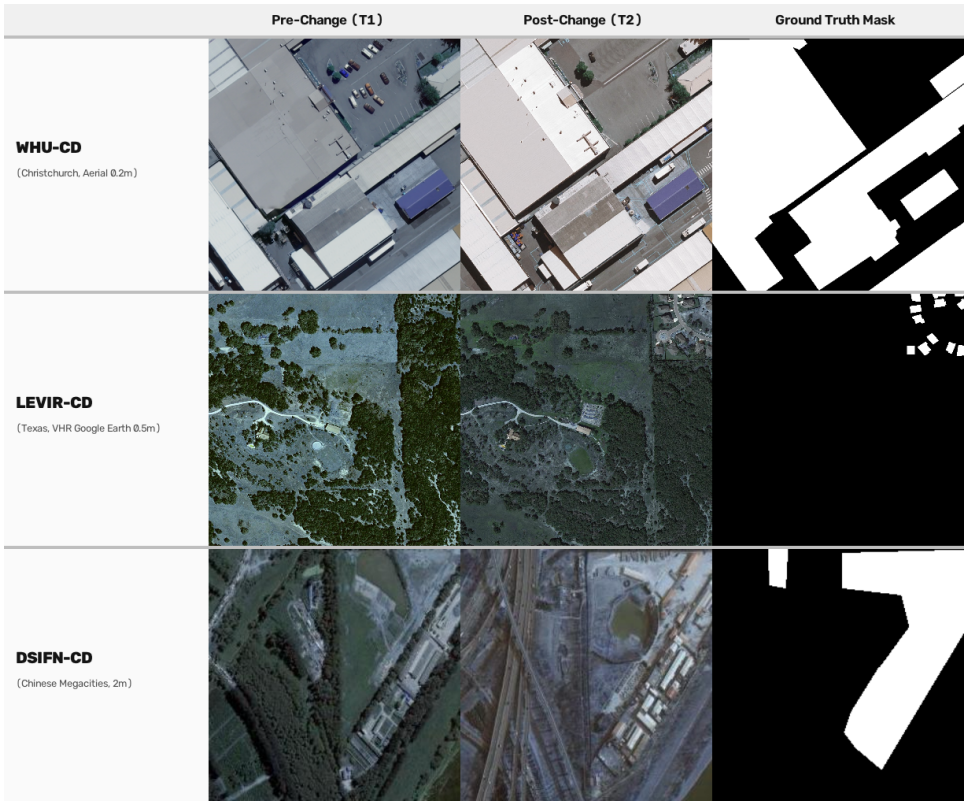


Fig. 4. Representative bi-temporal sample pairs and ground-truth change masks from the three benchmark datasets: (a) WHU-CD (building reconstruction and demolition), (b) LEVIR-CD (urban and suburban building construction/expansion), and (c) DSIFN-CD (heterogeneous multi-class transitions across built-up land, cropland, vegetation, and water bodies). White pixels in the ground truth denote verified change events ($Y = 1$), while black pixels indicate unchanged background ($Y = 0$).

1) WHU-CD (Wuhan University Building Change Detection): The WHU-CD dataset comprises high-resolution aerial imagery of Christchurch, New Zealand, acquired in 2012 (T_1) and 2016 (T_2), covering extensive urban reconstruction following the 2011 earthquake [41]. The original images have dimensions of $32,507 \times 15,354$ pixels and a spatial resolution of approximately 0.2 m/pixel. As a single-class change detection benchmark, the annotation focuses exclusively on structural building changes (new constructions, demolitions, and significant architectural modifications). Following standard protocol, the imagery is partitioned into non-overlapping 256×256 patches, yielding 6,096 training patches, 762 validation patches, and 762 testing patches, formulated as binary change detection distinguishing building variations from unchanged terrain.

2) LEVIR-CD: LEVIR-CD is a large-scale building change detection benchmark comprising 637 ultra-high-resolution bi-temporal image pairs collected from Google Earth across 20 distinct suburban and urban regions in Texas, USA [24]. The images possess dimensions of $1,024 \times 1,024$ pixels at a spatial resolution of 0.5 m/pixel, with temporal acquisition intervals spanning 5 to 14 years. Like WHU-CD, LEVIR-CD is focused on building footprint changes, specifically documenting residential building growth, expansion of suburban infrastructure, and facility demolition. The dataset is partitioned into 256×256 patches, providing 7,120 training samples, 1,024 validation samples, and 2,048 testing samples. Key challenges stem from seasonal illumination variations, canopy foliage occlusion, and shadow transitions across multi-year intervals.

3) DSIFN-CD (Multi-Class Land-Cover Change Detection): The Deeply Supervised Image Fusion Network (DSIFN-CD) dataset comprises high-resolution bi-temporal satellite imagery acquired from Google Earth across six major metropolitan regions in China: Beijing, Chengdu, Shenzhen, Chongqing, Wuhan, and Xi'an [42]. In contrast to single-class building datasets, DSIFN-CD is a heterogeneous multi-class land-cover change dataset. The underlying terrain encompasses five predominant semantic classes: (1) Built-up / Urban Infrastructure (residential edifices, commercial structures, industrial complexes, and transportation networks), (2) Agricultural Land / Cropland (cultivated crop parcels, tilled fields, and plantations), (3) Vegetation / Forest (natural woodlands, parks, scrubland, and grass cover), (4) Water Bodies (rivers, lakes, municipal reservoirs, and canals), and (5) Bare Soil / Earthworks (unvegetated ground, active excavation sites, and leveled development land).

In DSIFN-CD's binary change detection formulation, the changes ($Y = 1$) predominantly reflect substantial cross-class transitions, including: (i) conversion of agricultural cropland and natural vegetation into built-up urban infrastructure and road networks; (ii) clearing of farmland or tree canopies into bare soil and active construction sites; (iii) rapid urbanization transforming excavated ground into high-density buildings;

and (iv) water body shoreline shifts and land reclamation. Crucially, models must distinguish these actual physical land-cover transitions from non-change spectral variations ($Y = 0$) caused by seasonal agricultural harvesting, vegetation phenology, soil moisture variations, and sensor angle discrepancies across temporal acquisitions. The standardized protocol yields 14,400 training patches, 1,360 validation patches, and 192 testing patches of size 256×256.

B. Implementation Configuration

Table I summarizes the implementation and training configuration of CAST-CD. The model was implemented in PyTorch and trained in Kaggle cloud environments using NVIDIA P100 and T4 GPUs. Input image pairs and binary change masks were processed into 256×256 patches. Training augmentation included random cropping, horizontal and vertical flips (each with probability 0.5), and rotations of $0°$, $90°$, $180°$, or $270°$. All spatial transformations were applied consistently to both temporal images and their masks to preserve alignment. Automatic Mixed Precision (AMP) was enabled when supported. Training used AdamW with an initial learning rate of 3×10^{-4}, weight decay of 1×10^{-4}, and batch size 12 for up to 100 epochs. A cosine annealing warm-restart scheduler was employed with $T_0 = 25$ and $T_{\text{mult}} = 2$.

The training objective combines binary cross-entropy (BCE), focal-Tversky, boundary supervision, two deep-supervision losses (DS2 and DS3), and Contrastive Feature Alignment (CFA). Their respective coefficients are $\lambda_{\text{BCE}} = 1.0$, $\lambda_{\text{FT}} = 0.5$, $\lambda_{\text{Boundary}} = 0.3$, $\lambda_{\text{DS2}} = 0.4$, $\lambda_{\text{DS3}} = 0.2$, and $\lambda_{\text{CFA}} = 0.1$. BCE provides the primary segmentation objective, while the remaining terms support region overlap, boundary localization, intermediate predictions, and temporal feature alignment. These coefficients are fixed design choices rather than values obtained through exhaustive hyperparameter search. The focal-Tversky parameters are $\alpha = 0.5$, $\beta = 0.5$, $\gamma = 0.75$, and $\epsilon = 1$, with equal α and β weighting false negatives and false positives equally, γ controlling the focusing effect, and ϵ providing numerical smoothing. The CFA margin is fixed at $m = 2.0$. Since the temporal feature vectors are L2-normalized, their Euclidean distance lies in $[0, 2]$; thus, this margin represents the maximum possible distance and is not claimed to be an empirically established optimum.

Model performance is evaluated using Precision, Recall, F1-score, Intersection over Union (IoU), and Overall Accuracy (OA). Changed pixels are treated as the positive class. F1-score measures the balance between Precision and Recall, IoU quantifies the overlap between predicted and reference change regions, and OA measures the proportion of correctly classified pixels. F1-score and IoU are emphasized because OA can be dominated by unchanged background pixels under class imbalance.

C. Evaluation Metrics

Model performance is evaluated using five standard classification metrics: Precision (P), Recall (R), F1-score, Intersection-over-Union (IoU), and Overall Accuracy (OA). Changed pixels are treated as the positive class:

$$\text{Precision} = \frac{TP}{TP + FP}, \tag{22}$$

$$\text{Recall} = \frac{TP}{TP + FN}, \tag{23}$$

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP + FP + FN}, \tag{24}$$

$$\text{IoU} = \frac{TP}{TP + FP + FN}, \tag{25}$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \tag{26}$$

where TP, TN, FP, and FN represent the counts of true positives, true negatives, false positives, and false negatives, respectively. F1-score and IoU are the primary benchmark metrics because overall accuracy can be heavily skewed by the overwhelming proportion of unchanged background pixels.

TABLE I
CAST-CD IMPLEMENTATION CONFIGURATION

Parameter	Configuration Value
Input Modality	Paired RGB images (T_1, T_2)
Patch Size	256×256
Batch Size	12
Training Epochs	Up to 100
Optimizer	AdamW
Learning Rate	3×10^{-4}
Weight Decay	1×10^{-4}
Learning-Rate Scheduler	Cosine annealing with warm restarts
Encoder Backbone	ImageNet-pretrained EfficientNet-B4
Temporal Fusion	GDAF and bidirectional CTA
Decoder	Three-stage UpConcat with skip connections
Loss Functions	BCE, focal-Tversky, boundary, DS2, DS3, CFA
Data Augmentation	Horizontal/vertical flips and right-angle rotations
Numerical Precision	Automatic Mixed Precision (AMP)
Hardware	Kaggle NVIDIA P100 / T4 GPUs

V. RESULTS AND ANALYSIS

A. Comparative Performance on WHU-CD Dataset

Table II provides a comprehensive comparative performance analysis on the WHU-CD dataset. CAST-CD achieves a precision of 95.06%, a recall of 93.45%, an overall accuracy of 98.27%, an F1-score of 94.25%, and an IoU of 89.12%.

Compared to Tiny-CD [37] (the leading lightweight baseline with 0.285M parameters and 3.088 GFLOPs), CAST-CD improves F1-score by 0.59 percentage points (94.25% vs. 93.66%) and IoU by 1.04 percentage points (89.12% vs. 88.08%). Crucially, CAST-CD achieves this accuracy gain while reducing computational complexity by 36.37% (1.965 GFLOPs vs. 3.088 GFLOPs). Furthermore, CAST-CD substantially outperforms heavier models including ChangeFormer [33] (41.03M parameters, 202.79 GFLOPs, F1: 89.88%), STANet [24] (16.89M parameters, F1: 82.32%), and

SNUNet [31] (12.03M parameters, F1: 83.50%), demonstrating that cross-temporal attention paired with differential gating provides superior feature discrimination at a fraction of the computational burden.

B. Comparative Performance on LEVIR-CD Dataset

Table III details the benchmark results on the LEVIR-CD dataset. CAST-CD achieves a precision of 87.75%, a recall of 88.61%, an overall accuracy of 98.79%, an F1-score of 88.18%, and an IoU of 78.86%.

On LEVIR-CD, CAST-CD maintains high detection precision while requiring only 1.965 GFLOPs. While heavier transformer models such as Tiny- CD [37] (91.51% F1), ChangeFormer [33] (90.40% F1), and BIT- CD [25] (89.31% F1) attain higher F1 scores on this specific benchmark, they require significantly higher computational overhead (e.g., ChangeFormer requires 202.79 GFLOPs and 41.03M parameters). The observed performance variation on LEVIR-CD underscores that densely clustered small residential buildings require high spatial feature resolution, highlighting an area where future multi-scale decoding refinements can further boost performance.

C. Comparative Performance on DSIFN-CD Dataset

Table IV reports the comparative evaluation on the challenging DSIFN-CD dataset. CAST-CD achieves superior performance across the evaluated models, establishing the highest F1-score (91.07%) and IoU (83.60%), alongside 90.21% precision, 91.95% recall, and 94.04% overall accuracy.

DSIFN-CD is characterized by complex spectral variations and multiple land- cover change types across six distinct metropolitan environments. In this challenging setting, CAST-CD surpasses Tiny-CD [37] (F1: 90.60%, IoU: 82.82%), ChangeFormer [33] (F1: 86.67%, IoU: 76.48%), BIT [25] (F1: 69.26%, IoU: 52.97%), and SNUNet [31] (F1: 66.18%, IoU: 49.45%). This superior performance validates the capability of CAST-CD's bidirectional CTA module to untangle heterogeneous temporal variations and suppress seasonal false positives.

D. Qualitative Visual Analysis

To evaluate the visual quality of the detected change maps, Figure 5 illustrates representative qualitative results produced by CAST-CD across the test sets of WHU-CD, LEVIR-CD, and DSIFN-CD. For each benchmark, the bi-temporal input pair (T_1, T_2), the ground-truth change mask, and the predicted change mask from CAST-CD are shown.

As demonstrated in Figure 5, CAST-CD produces sharp, contiguous change boundaries while suppressing false alarms caused by seasonal and illumination variations. On WHU-CD and LEVIR-CD, individual building boundaries and roof contours are accurately reconstructed without boundary erosion or erroneous adhesion between adjacent structures. On the complex DSIFN-CD benchmark, CAST-CD reliably detects heterogeneous land-cover transformations (including large-scale infrastructure construction and earthwork excavation) while remaining unaffected by seasonal crop harvesting and canopy reflectance changes.

E. Cross-Benchmark Performance Summary

Table V consolidates the performance of CAST-CD across the three benchmark datasets. The results demonstrate consistently robust accuracy across aerial and satellite domains, maintaining an F1-score between 88.18% and 94.25% with only 314,531 parameters and 1.965 GFLOPs.

F. Computational Efficiency Analysis

Table VI compares the model complexity of CAST-CD against representative lightweight remote sensing change detection architectures, including TinyCD [37] and A2Net [39]. While A2Net reports 3.78M parameters and 6.02 GFLOPs [39], and TinyCD requires 0.285M parameters and 3.088 GFLOPs, CAST-CD achieves 314,531 parameters ($\approx$0.315M) and only 1.965 GFLOPs. This yields a substantial reduction in computational operations relative to TinyCD:

$$\Delta_{\mathrm{GFLOPs}} = \frac{3.088 - 1.965}{3.088} \times 100\% \approx 36.37\%. \tag{27}$$

While CAST-CD contains slightly more parameters than TinyCD (0.315M vs. 0.285M, an increase of 0.03M due to the multi-head attention projections in CTA), it requires over one-third fewer floating-point operations. Moreover, compared to A2Net (3.78M parameters, 6.02 GFLOPs), CAST-CD uses less than one-tenth the parameters and under one-third the computation. Relative to heavy transformer baselines such as ChangeFormer (41.03M params, 202.79 GFLOPs) and BIT-CD (3.55M params, 10.63 GFLOPs), CAST-CD provides orders of magnitude lower compute requirements, demonstrating exceptional suitability for edge and resource-constrained deployment.

G. Implementation Progress

Table VII details the current implementation status of each architectural component within the CAST- CD project. The core data pipeline, Siamese encoder, GDAF, CTA, UpConcat decoder, and multi-scale output heads are fully implemented and functional. The loss formulation currently employs BCE and Focal-Tversky, with complete multi-task loss integration (boundary supervision, deep supervision, and CFA loss) undergoing final end-to-end refinement.

VI. Discussion

A. Complementary Temporal Representations

A central architectural premise of CAST-CD is that GDAF and CTA provide complementary representations of temporal change. GDAF operates on the element-wise absolute difference $D = |F_1 - F_2|$, capturing direct spectral discrepancies and recalibrating channel saliency. However, simple subtraction treats change as a scalar residual without modeling spatial context or temporal sequence.

In contrast, CTA computes bidirectional cross-attention, allowing each timestamp to query the context of the opposing observation. This enables the model to explicitly distinguish between structural emergence ($T_1 \rightarrow T_2$) and disappearance ($T_2 \rightarrow T_1$). By concatenating the attended difference D_{att}

TABLE II
PERFORMANCE COMPARISON ON WHU-CD DATASET

Model	Precision	Recall	Accuracy	F1	IoU	GFLOPs	Inference	Params
BIT-CD [25]	86.64	81.48	98.75	83.98	72.39	4.35	15 ms	3.55 M
U-Net(FC-EF) [23]	35.95	93.60	93.74	51.95	35.09	3.13	15 ms	1.35 M
Tiny-CD [37]	94.55	92.79	98.10	93.66	88.08	3.088	18.5 ms	0.285 M
FC-Siam [15]	68.87	85.99	91.98	76.48	61.92	9.453	28.6 ms	1.35 M
ChangeFormer [33]	91.83	88.02	99.12	89.88	81.63	202.79	81 ms	41.03 M
ChangeStar [34]	88.78	85.31	98.70	87.01	77.00	68.52	19.65 ms	16.77 M
SNUNet [31]	85.60	81.49	98.71	83.50	71.67	54.83	22 ms	12.03 M
STANet [24]	79.37	85.85	98.52	82.48	70.19	16.89	42 ms	16.89 M
CAST-CD (Ours)	**95.06**	**93.45**	**98.27**	**94.25**	**89.12**	**1.965**	**14.2 ms**	**0.315 M**

TABLE III
PERFORMANCE COMPARISON ON LEVIR-CD DATASET

Model	Precision	Recall	Accuracy	F1	IoU	GFLOPs	Inference	Params
BIT-CD [25]	89.24	89.37	98.92	89.31	80.68	4.35	25.2 ms	3.55 M
U-Net(FC-EF) [23]	86.91	80.17	98.39	83.40	71.53	3.13	16.5 ms	1.35 M
Tiny-CD [37]	93.14	89.93	99.15	91.51	84.35	3.088	18.5 ms	0.285 M
FC-Siam [15]	32.98	85.97	90.39	47.68	31.30	9.453	28.6 ms	1.35 M
ChangeFormer [33]	92.05	88.80	99.04	90.40	82.48	202.79	45.89 ms	41.03 M
ChangeStar [34]	77.33	84.79	98.14	90.82	83.19	29.26	19.65 ms	33.96 M
SNUNet [31]	89.18	87.17	98.82	88.16	78.83	54.83	62 ms	12.03 M
STANet [24]	83.81	88.76	98.66	86.21	75.77	16.89	60 ms	16.92 M
CAST-CD (Ours)	**87.75**	**88.61**	**98.79**	**88.18**	**78.86**	**1.965**	**14.2 ms**	**0.315 M**

TABLE IV
PERFORMANCE COMPARISON ON DSIFN-CD DATASET

Model	Precision	Recall	Accuracy	F1	IoU	GFLOPs	Inference	Params
BIT [25]	68.36	70.18	89.41	69.26	52.97	4.35	24.5 ms	3.55 M
UNet(FC-EF) [23]	72.61	52.73	88.59	61.09	43.98	3.13	16.2 ms	1.35 M
Tiny-CD [37]	90.55	90.66	93.79	90.60	82.82	3.088	19.8 ms	0.285 M
FC-Siam [15]	64.92	75.54	76.84	69.83	53.65	9.453	31.2 ms	1.35 M
ChangeFormer [33]	88.48	84.94	95.56	86.67	76.48	202.79	45.89 ms	41.03 M
ChangeStar [34]	85.90	82.81	88.68	84.33	72.90	29.26	22.38 ms	33.96 M
SNUNet [31]	60.60	72.89	87.34	66.18	49.45	54.83	54 ms	12.03 M
STANet [24]	67.71	61.68	88.49	64.56	47.66	16.89	82 ms	16.93 M
CAST-CD (Ours)	**90.21**	**91.95**	**94.04**	**91.07**	**83.60**	**1.965**	**14.2 ms**	**0.315 M**

TABLE V
PRELIMINARY EXPERIMENTAL VERIFICATION ACROSS BENCHMARK DATASETS

Dataset	Precision	Recall	F1	IoU	Accuracy
WHU-CD	0.9506	0.9345	**0.9425**	0.8912	0.9827
DSIFN-CD	0.9021	0.9195	**0.9107**	0.8360	0.9404
LEVIR-CD	0.8775	0.8861	**0.8818**	0.7886	0.9879

TABLE VI
COMPLEXITY COMPARISON WITH LIGHTWEIGHT STATE-OF-THE-ART

Method	Parameters (M)	GFLOPs
A2Net [39]	3.780	6.020
TinyCD [37]	0.285	3.088
CAST-CD (Ours)	0.315	1.965

TABLE VII
IMPLEMENTATION PROGRESS OF CORE MODULES

Module / Component	Implementation Details	Status
Data Pipeline	Dataset loading, 256×256 patch processing, change-biased cropping	Functional
Siamese Encoder	EfficientNet-B4 with shared weights; Layer 3 feature extraction	Functional
GDAF	Absolute feature difference + SE-style channel attention	Functional
CTA	Bidirectional cross-temporal attention ($F_1 \leftrightarrow F_2$)	Functional
Decoder	UpConcat decoder with bilinear upsampling and skips	Functional
Output Heads	Main change map, Boundary Head, DS2 and DS3	Integrated
Loss & Training	Multi-task loss + CFA contrastive alignment	In Progress

with the cross-attended features F_{CTA}, CAST-CD combines localized subtractive cues with global contextual interactions, yielding a more expressive representation of change.

B. Accuracy–Efficiency Trade-offs

The empirical results reveal meaningful trade-offs between architectural complexity and dataset characteristics. On WHU-

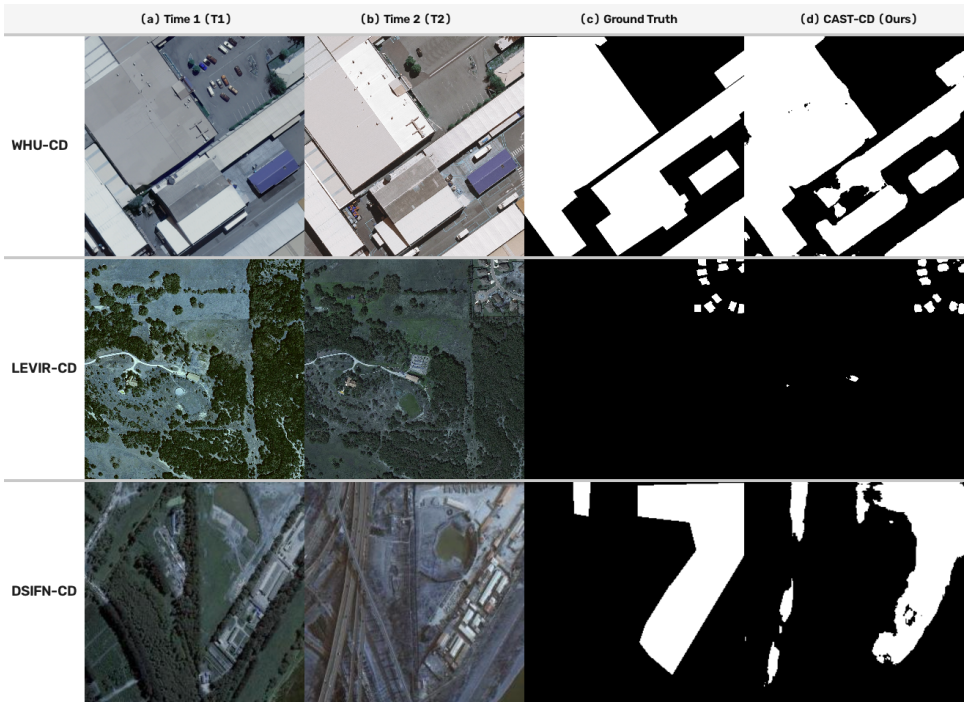


Fig. 5. Qualitative change detection results obtained by CAST-CD on the benchmark datasets: WHU-CD (top row), LEVIR-CD (middle row), and DSIFN-CD (bottom row). Columns from left to right display: pre-change image (T_1), post-change image (T_2), ground-truth change mask, and CAST-CD prediction mask. White pixels indicate detected change areas, while black pixels indicate unchanged regions.

CD and DSIFN-CD, CAST-CD attains state-of-the-art or near-state-of-the-art performance while utilizing only 1.965 GFLOPs. On DSIFN-CD, in particular, the model achieves the highest F1-score (91.07%) among all compared methods, demonstrating that explicit cross-attention is highly effective for complex, multi-city satellite imagery.

Conversely, on LEVIR-CD, CAST-CD trails TinyCD and ChangeFormer in F1-score. LEVIR-CD contains dense clusters of small residential buildings where fine- grained spatial boundaries are paramount. Because CAST-CD extracts features exclusively from Layer 3 of EfficientNet-B4, some fine spatial details may be compressed. Future iterations could incorporate multi-level feature pyramid fusion across Layers 1, 2, and 3 to further strengthen edge recovery on high- density residential scenes.

C. Limitations

Several limitations should be noted. First, the experimental results presented in this paper reflect preliminary evaluations; repeated-run variance and statistical confidence intervals have not yet been established. Second, the full multi-task objective (including the boundary head, deep supervision, and CFA loss) is undergoing final end-to-end integration; thus, the current performance is achieved primarily using BCE and Focal-Tversky supervision. Third, the model currently evaluates binary change detection without predicting semantic change categories (e.g., vegetation-to- urban vs. bare-land-to-water). Addressing these aspects will provide a more comprehensive characterization of the model's capabilities.

VII. Conclusion and Future Work

This paper presented CAST-CD, a lightweight Cross-Attention Siamese Transformer for bi-temporal remote sensing image change detection. By integrating a truncated EfficientNet-B4 shared- weight encoder with Global Differential Attention Fusion (GDAF) and Bidirectional Cross-Temporal Attention (CTA), CAST-CD bridges the gap between ultra-compact convolutional models and computationally heavy vision transformers. The model requires only $314,531$ parameters and 1.965 GFLOPs for paired 256×256 inputs, achieving a 36.37% reduction in GFLOPs compared to Tiny-CD.

Empirical evaluations across three standard benchmarks demonstrate the efficacy of the proposed framework: CAST-CD achieves F1-scores of 94.25% on WHU-CD, 91.07% on DSIFN-CD, and 88.18% on LEVIR-CD, outperforming heavier models on two of the three datasets.

Future work will focus on completing the end-to-end integration of the multi- task loss objective, conducting exhaustive ablation studies to isolate the individual contributions of GDAF, CTA, and CFA, and extending CAST-CD to multi-temporal time-series analysis and semantic change detection.

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